

(Breaking) Intergenerational Transmission of Mental Health

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Outline

- 1 Motivation
- 2 Related Literature and Contribution
- 3 Setting and Data
- 4 Empirical Framework
- 5 Intergenerational Persistence in Health
- 6 Targeted Policy and Breaking the Cycle
- 7 Conclusion

The Global Mental Health Burden

- Mental health disorders affect **more than one billion people worldwide**
- Among leading causes of disability and rising share of global disease burden
- Economic costs are large and growing:
 - US: \$201 billion in 2013; projected \$225 billion by 2019
 - Norway: NOK32 billion (\$3.7B USD) in 2013; NOK65 billion by 2023
- COVID-19 has further exacerbated demand for mental health care
- True economic costs exceed medical spending: productivity loss, forgone taxes, externalities on others

An Understudied Externality: Intergenerational Transmission

- One key externality is the intergenerational association between parent and child mental health
- If mental health conditions transmit across generations, costs imposed on future generations are substantial and unaccounted for
- Three research questions:
 - What is the **association** between parent and child mental health?
 - How different is it from physical health? How important is the **extended family**?
 - Can a **targeted low-touch policy** break the intergenerational link?
- Answers require: population-level data, objective health measures, family linkages across generations

Preview of Main Findings

- **Persistence is large:** having a parent with a mental health diagnosis raises child's probability of mental health diagnosis by 9.3 percentage points (40% of control mean)
- **Dynasty matters:** focusing only on parents understates persistence by 42% — extended family adds substantial explanatory power
- **Mental $\approx$ physical:** intergenerational associations in mental and physical health are broadly similar in magnitude; genetics likely not the full story
- **Policy can help:** a 2007 Norwegian pilot program reduced the parent-child mental health association by 39%
 - Larger effects for children of college-educated parents; raises equity concerns

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Prior Literature on Intergenerational Health

- **General health persistence:** Andersen (2021), Björkegren et al. (2022), Halliday et al. (2021) — registry-based estimates tend to be smaller than survey-based
- **Specific conditions:** birth weight (Currie & Moretti 2003), cardiovascular (Lloyd-Jones et al. 2004), BMI (Classen 2010)
- **Mental health specifically is scarcer:**
 - Survey-based: Johnston, Schurer, & Shields (2013); Bencsik, Halliday, & Mazumder (2021); Vera-Toscano & Brown (2021)
 - Specific disorders: Akbulut-Yuksel & Kugler (2016) for depression; Eley et al. (2015) for anxiety
 - Almost all rely on **self-reported survey data**: recall bias, small samples, ordered intervals
- Survey estimates of parent-child mental health ICs: 0.13 (Johnston et al.) to 0.22 (Bencsik et al.)
- Only Andersen (2021) provides population-level estimates using administrative GP/hospitalization data for Denmark

This Paper's Contribution

- Connects three literatures:
 - ① **Intergenerational health persistence**: adds administrative mental health, dynasty component, causal policy variation
 - ② **Dynastic human capital** (Adermon, Lindahl & Palme 2021): replicates for Norway, extends beyond education to health outcomes
 - ③ **Health interventions in childhood**: Hjørst, Sølvesten, & Wüst (2017); Bütikofer & Salvanes (2020); Baranov et al. (2020) — light-touch policies can reduce persistent disadvantage
- Broader implication: standard program evaluations that stop at the directly treated generation **may understate returns** to mental health investments

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Why Norway?

- Healthcare is **highly subsidized and universally accessible** — limits selection into treatment based on income
- Unique **personal and family identifiers** across all administrative registers: link individuals across four generations
- General practitioners (GPs) and primary care ER doctors are **obligated to report all consultations**
 - Near-universal coverage for children 13–18: 91% visited primary healthcare before seeing a specialist
- Reduces concerns about underreporting relative to survey data:
 - Bharadwaj, Pai, and Suzie-delyte (2017): large underreporting of MH in surveys vs. administrative records
- Private insurance/clinics have negligible market share ($\approx 10,000$ adults with private coverage in 2003)

Data Sources

- **Family registers (1992–2015):** link parents, children, and extended family via birth certificates; construct extended family horizontally (siblings, cousins, spouses)
- **KUHR — Control and Payment of Health Refunds (2006–2020):** children's health
 - GP and primary care ER visits; ICPC-2 coded
 - Available from 2006 onward
- **Social Security sickness leave registry (1992+):** parental health
 - All absences > 3 days certified by a physician; ICPC-2 coded
- **Demographic, education, and income registers:** controls and heterogeneity analyses

Measuring Mental Health

Children's Mental Health:

- **Source:** KUHR registry — GP and ER visits between ages 13 and 18
- **Mental health indicator:** any GP or ER visit with an ICPC-2 code starting with the letter “P”

Parents' Mental Health:

- **Source:** Sickness leave registry — any physician-certified sickness leave for mental health reasons (ICPC-2 code “P”) at ages 25–30
- **Why this window:** ensures parental health is measured *before* the child's outcome window (ages 13–18)
- 7% of mothers and 4% of fathers take at least one MH sickness leave at ages 25–30
- Most common MH condition: depression (both genders)

Limitations of the Sickness Leave Measure

- **Eligibility constraint:** only individuals in the labor market qualify — 6% of 25–30 year olds ineligible
 - Ineligibility more common among foreign-born, lower-income individuals
 - Our estimates likely represent an **upper bound** for severe conditions (sickest may not work); a **lower bound** for mild conditions (mild cases may not take leave)
- **Severity truncation:** not all mental health episodes lead to leave — depression manageable by medication may not appear
- **Coverage overlap:** for 2006–2008, compare sickness leave to KUHR data: 23% of individuals with a MH event in KUHR took sickness leave
- **External validity:** parents with MH diagnoses are somewhat more likely to be parents than those without, suggesting our sample is not severely selected
- **Robustness:** control for sickness leave eligibility determinants (education, income, immigration status) — results unchanged

Constructing the Extended Family (Dynasty)

- Following Adermon, Lindahl, and Palme (2021): define the dynasty to include **five branches** of the extended family beyond parents:
 - ① **Parents' siblings** (sp) — 25% genes shared
 - ② **Spouses of parents' siblings** (ssp) — plausibly unrelated genetically
 - ③ **Siblings of spouses of parents' siblings** (sssp) — plausibly unrelated
 - ④ **Parents' cousins** (cp) — 12.5% genes shared
 - ⑤ **Spouses of parents' cousins** (scp) — plausibly unrelated
- Associations for “plausibly unrelated” members (ssp, sssp, scp) constitute 18–15% of total intergenerational persistence $\Rightarrow$ genetics alone cannot explain the full picture
- Control for **number of members** in each branch of extended family to avoid mechanical effects

Sample Selection and Descriptive Statistics

- Start: all children born in Norway 1988–2007, resident ages 13–18 in 2006–2020 ($N = 732,437$)
- Restrict to:
 - Parents with labor income at ages 25–30 (sickness leave eligibility): drop 4,907
 - Children with known grandparents and great-grandparents: $N = 370,498$ preferred sample
- **Children:** 24% have any MH diagnosis; 6.96% have depression; 51% male
- **Mothers:** 54% take at least one sickness leave ages 25–30; 7% for MH reasons; most common reason: musculoskeletal (23%)
- **Fathers:** 28% take at least one sickness leave; 4% for MH reasons
- Parent-child MH correlations below 0.04 in the raw data (Pearson) — associations are real but moderate

Sample Statistics

	All Children		Analysis Sample	
	Mean	SD	Mean	SD
<i>Panel A: Children</i>				
Any MH diagnosis	23.09	42.14	24.00	42.71
Depression	6.72	25.03	6.96	25.45
Any specialist care (MH)	4.03	19.67	4.11	19.85
Male	0.51	0.50	0.51	0.50
GPA	4.09	0.82	4.05	0.83
≥ 1 parent college	0.36	0.48	0.33	0.47
<i>Panel B: Mother</i>				
Any sick leave (25–30)	0.50	0.50	0.54	0.50
Any MH sick leave (25–30)	0.06	0.24	0.07	0.26
Any depression SL (25–30)	0.04	0.20	0.05	0.21
<i>Panel C: Father</i>				
Any sick leave (25–30)	0.26	0.44	0.28	0.45
Any MH sick leave (25–30)	0.03	0.18	0.04	0.19
Any depression SL (25–30)	0.02	0.14	0.02	0.14

Children's health measured ages 13–18; parental health at ages 25–30. $N = 370,498$ (analysis sample).

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Econometric Model: Intergenerational Persistence

Regress child mental health on parental and extended family mental health:

$$Y_{it} = \alpha + \beta_p Y_{t-1}^p + \sum_{k \in \{sp, cp, ssp, sssp, scp\}} \beta_k Y_{t-1}^k + \gamma \mathbf{X}_{it} + \varepsilon_{it}$$

- Y_{it} : indicator (extensive) or count (intensive) of child's MH events, ages 13–18, $\times 100$
- Y_{t-1}^p : parental MH sickness leave, ages 25–30 (measured before child outcome)
- Y_{t-1}^k : MH sickness leave of extended family branch k
- $\mathbf{X}_{it}$: year of birth FE, year first observed in KUHR, gender, number of individuals per dynasty branch; *select specs* add SES controls
- Parameters of interest: β_p (parent-child) and β_k (extended family)
- Standard errors: Eicker-Huber-White, clustered at the family level

Identification: What We Can and Cannot Claim

- **What the design delivers:** associations / correlations between parent and child MH, not causal effects of parental MH on child MH
- These associations are policy-relevant regardless of mechanism:
 - Mental health diagnoses trigger initial treatment and follow-up costs
 - They motivate the targeted intervention in Section V
- **Timing:** parental health measured at ages 25–30, strictly before child's outcome window (ages 13–18) $\Rightarrow$ rules out co-diagnosis
- **Not causal but informative:** associations are robust to controlling for SES, school FE, GP FE, prenatal health, and birth characteristics
- Cannot separate genetic from social/environmental channels — dynasty analysis provides suggestive evidence that social factors matter

A Note on Measurement of Parental Health

- **Upper vs. lower bound concern:**
 - If milder conditions have lower persistence $\Rightarrow$ sicker individuals more likely captured $\Rightarrow$ estimates are an **upper bound**
 - If sickest individuals exit the labor market $\Rightarrow$ most severe cases missed $\Rightarrow$ estimates are a **lower bound**
- Authors conclude: upper bound concern is more plausible given the data
- **Validate** with 2006–2008 overlap of KUHR and sickness leave: 23% of individuals with a MH symptom in KUHR took sickness leave
- **Administrative vs. survey:** Bharadwaj et al. (2017) show large underreporting of MH in surveys — administrative records likely more accurate

Dynastic Correlations in Mental Health (Extensive Margin)

	(1) Parents only	(2) + Siblings	(3) Full dynasty	(4) + Physical health
Parents' MH	9.805*** (0.254)	9.664*** (0.254)	9.569*** (0.254)	9.257*** (0.255)
Parents' siblings MH		2.990*** (0.208)	2.848*** (0.208)	2.557*** (0.210)
Spouses of parents' siblings MH			1.417*** (0.256)	1.181*** (0.258)
Parents' cousins MH			1.654*** (0.163)	1.475*** (0.164)
Spouses of parents' cousins MH			0.676*** (0.195)	0.556*** (0.195)
Siblings of spouses of parents' siblings MH			1.025*** (0.206)	0.883*** (0.208)
Control for physical health	No	No	No	Yes
Control mean			22.9	
Sum of coefficients	9.8	12.7	17.2	15.9
R^2	0.052	0.052	0.053	0.053
N			370,498	

Dynastic Correlations in Mental Health: Intensive Margin

	(1) Parents only	(2) + Siblings	(3) Full dynasty	(4) + Physical health
Parents' MH	0.050*** (0.003)	0.050*** (0.003)	0.049*** (0.003)	0.048*** (0.003)
Parents' siblings MH		0.017*** (0.002)	0.017*** (0.002)	0.016*** (0.002)
Sp. of par. siblings MH			0.007*** (0.002)	0.006*** (0.002)
Parents' cousins MH			0.007*** (0.002)	0.006*** (0.002)
Sp. of par. cousins MH			0.002 (0.002)	0.002 (0.002)
Sib. of sp. of par. siblings MH			0.004** (0.002)	0.003* (0.002)
Control for physical health	No	No	No	Yes
Sum of coefficients	0.050	0.067	0.085	0.080
R^2	0.032	0.032	0.032	0.034
N			370,498	

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Main Results: Extensive Margin (Mental Health)

- **Parent-child association (Column 1):** 9.8 percentage points
 - Control group mean = 23% $\Rightarrow$ this is a **40% increase** in the probability of a MH event
- Adding extended family branches: parent coefficient barely changes
 - Parent-child association is largely **orthogonal** to extended family associations
- **Full dynastic model (Column 3):** parent coefficient = 9.6 pp; sum of all extended family = 7.6 pp $\Rightarrow$ dynasty adds 79% more
- **Controlling for physical health (Column 4):** parent coefficient = 9.3 pp — MH associations not driven by cross-condition correlations
- All extended family branches are statistically significant; magnitude decreasing in familial distance

Main Results: Intensive Margin

- Substitute indicator variables with **standardized counts** of MH health events (children) and MH sickness leave days (parents and extended family)
- **Parent-child association:** 0.05 standard deviations (extensive: ≈ 0.05 too) — broadly comparable
- **Extended family:** sum of coefficients = 0.037, or 75.5% of the parent-child association
 - Very similar ratio to the extensive margin
- Spouses of parents' cousins (scp) becomes statistically insignificant in the intensive margin saturated model
- Physical health intensive margin: sum of extended family = 76.3% of parent-child association — nearly identical to mental health
- Intergenerational health elasticities (intensive margin): **0.05 for MH**, 0.06 for physical health
 - Grows to 0.09–0.10 when dynasty is included
 - Substantially smaller than socioeconomic outcomes (income 0.34, wealth 0.35, education 0.25)

Dynastic Effects: Why the Extended Family Matters

- Ignoring dynasty **underestimates** total intergenerational persistence by 42%
- Extended family associations decrease with genetic distance:
 - Parents' siblings (sp): 14.8% of parent-child association
 - Spouses of parents' siblings (ssp): 7.1% — plausibly unrelated genetically
 - Spouses of parents' siblings and their siblings: smallest, but still significant
- “Plausibly unrelated” members (ssp, sssp, scp) account for 18% of total extensive MH persistence
 - Suggests social mechanisms (modeling behavior, resources, awareness) matter — not purely genetic
- Contrast with **educational persistence**: dynastic educational associations constitute only 6% of the corresponding parent-child association
- Mental health is more “visible” within extended families than education — may enable social transmission

Physical vs. Mental Health Persistence

- **Prior literature** often assumes mental $\neq$ physical health in intergenerational persistence (different mechanisms, heritability)
- **This paper's finding:** MH and physical health (OH) associations are largely similar
 - Extensive margin: parent-child MH IC = 0.050, OH IC = 0.062 — modestly larger for physical health
 - Intensive margin: nearly identical magnitudes
- MH associations are unaffected by controlling for physical health of generation $t - 1$ (and vice versa)
- Extended family patterns mirror each other: parent-child association independent of dynasty for both MH and OH
- Implication: the mechanisms driving intergenerational health persistence — genetic, social, environmental — are not fundamentally different across MH and OH

Comparing Health and Educational Persistence

- Replicate Adermon, Lindahl, and Palme (2021) for Norway using Grade 10 GPA as outcome
- **Parent-child education elasticity:** 0.45 — nine times larger than MH (0.05)
- **Dynasty sum for education:** 0.549 in Norway (vs. 0.518 in Sweden)
- Dynastic education associations **do not change** even when controlling for dynastic health associations
 - Education and health channels are separate — health does not mediate educational transmission
- “Genetically unrelated” dynastic members matter more for health (18%) than for education (6%)
 - Mental and physical health issues may be more “visible” within extended families than education, enabling social transmission

Magnitudes Relative to Prior Literature

- Administrative data estimates tend to be **two to four times smaller** than survey estimates
 - U.K. survey ICs: 0.13–0.22 (Johnston et al.; Bencsik et al.)
 - This paper (Norway, admin): parent-child IC ≈ 0.05 (intensive margin)
- Consistent with other registry-based work:
 - Björkegren et al. (2022): 0.11–0.14 using hospitalizations in Sweden
 - Andersen (2021): 0.09–0.14 for general health in Denmark
- Likely explanations for gap vs. surveys:
 - Upward bias from recall, subjective perception, Likert scale in survey data
 - Norway's universal healthcare reduces selection into treatment
- Norway is not an outlier: OECD data suggest comparable MH burden to other developed countries

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The 2007 Pilot Program: Modellkommuneforsøket

- In 2007, the Norwegian Directorate for Children, Youth, and Family Affairs launched a **pilot in 26 municipalities**
- Goal: find best practices for supporting young children (ages 0–6) of parents with mental health conditions
- Municipalities received NOK100,000–290,000 annually from central government
- The program was **light-touch and varied** across municipalities:
 - Screening tools for psychological distress among targeted children
 - Establishing specialist teams and interagency cooperation
 - Childcare center psychologists; home visits; parenting programs (PMTO)
 - Family coordination and case managers
- By 2010, all pilot municipalities had a registration system and screening tools
- Success led to a **2010 national law** mandating similar practices across all Norwegian municipalities

Empirical Strategy: Triple Differences

Since 26 out of 428 municipalities were not chosen randomly, use a **triple differences** design:

$$Y_{icm} = \alpha + \beta(E_{ic} \times D_{im} \times MPH_i) + \delta_1 MPH_i + \delta_2(E_{ic} \times MPH_i) + \delta_3(D_{im} \times MPH_i) + \delta_4(E_{ic} \times D_{im}) + \mathbf{X}'_i \boldsymbol{\mu} + \rho_c + \pi_m + \varepsilon_{icm}$$

- Y_{icm} : any MH GP/ER visit ages 13–18 for child i , cohort c , municipality m ($\times 100$)
- MPH_i : indicator for parental MH diagnosis (2000–2010)
- D_{im} : indicator for treated municipality
- E_{ic} : indicator for child aged ≤ 6 in 2007 (eligible cohort)
- β : ITT estimate of how the program affects the parent-child MH association
- Controls: year of birth FE, municipality FE, child gender, parent ages, birth weight, birth rank, parental education
- Municipalities matched via Mahalanobis distance on pre-intervention characteristics

Credibility: Parallel Trends and Placebo Tests

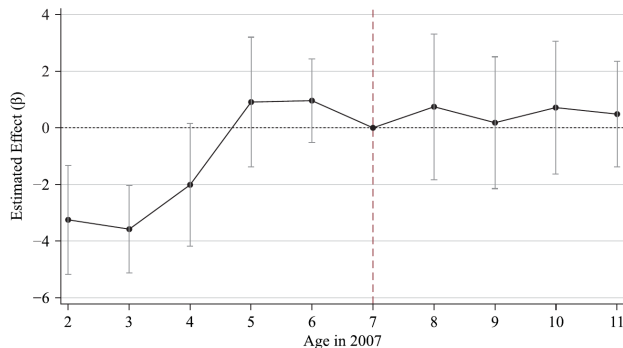


Figure 1

Effects of the Intervention by Age in 2007

Notes: This figure uses the same sample and outcome as Table 9. The figure presents estimates for β^r from Equation 4. Whiskers show 95 percent confidence intervals for each point estimate. Standard errors are clustered by children's municipality of birth.

Policy Effects on the Intergenerational Association

- **Baseline parent-child association** (control municipalities, ineligible cohort): 9.7 pp
 - Very close to 9.8 pp in the full-population analysis
- **Treatment effect ($\hat{\beta}$):** -3.8 pp (statistically significant at 1%)
 - Program **reduced the parent-child mental health association by 39%**
- Robust to intensive margin transformation (Panel B): -1.1 sickness leave days per year
- Robust to controlling for municipal trends in health services (Panel C)
- Most affected: youngest children at the start of the program (ages 2–4 in 2007)
 - Longer exposure and earlier age $\Rightarrow$ larger effects

Effects of the Pilot Program on Intergenerational Persistence

	All (1)	College (2)	No College (3)	Males (4)	Females (5)
<i>Panel A: Baseline</i>					
1[Parental MH]	9.730*** (0.461)	8.983*** (0.532)	10.104*** (0.671)	8.699*** (0.669)	10.840*** (0.905)
1[Parental MH] $\times$ (Age ₂₀₀₇ $\leq$ 6) $\times$ Pilot	-3.796*** (0.927)	-6.210*** (1.307)	-2.141 (1.390)	-3.603** (1.574)	-3.916*** (1.253)
<i>Panel B: Duration</i>					
1[Parental MH]	9.801*** (0.466)	8.965*** (0.545)	10.148*** (0.656)	8.649*** (0.730)	11.036*** (0.943)
1[Parental MH] $\times$ Duration $\times$ Pilot	-1.114*** (0.277)	-2.425*** (0.488)	-0.568 (0.341)	-1.403*** (0.507)	-0.795* (0.403)
<i>Panel C: Municipality trends</i>					
1[Parental MH]	9.703*** (0.457)	9.007*** (0.535)	10.076*** (0.671)	8.673*** (0.667)	10.804*** (0.887)
1[Parental MH] $\times$ (Age ₂₀₀₇ $\leq$ 6) $\times$ Pilot	-3.778*** (0.920)	-6.145*** (1.327)	-2.137 (1.388)	-3.610** (1.558)	-3.866*** (1.242)
N	129,683	60,596	69,087	66,074	63,609

Heterogeneity: Who Benefits Most?

- **By parental education:**
 - College-educated: -6.2 pp treatment effect (significant at 1%)
 - Non-college: -2.1 pp (statistically insignificant)
- Pilot was more effective for children of college-educated parents
 - Possibly: better ability to navigate and utilize offered services
 - Policy implication: **reduces average** parent-child MH association but **increases within-generation inequality** in mental health outcomes
- **By child gender:** broadly similar effects for boys and girls on extensive margin; 77% larger for boys on intensive margin
 - Boys may be more responsive to positive inputs, especially in childhood (Autor et al. 2019)
- Authors flag this as a **cautionary tale** for policymakers: interventions that reduce average persistence can still widen inequality within the affected generation

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Main Takeaways

- **Persistence is substantial:** parental MH diagnosis raises child MH probability by 9.3 pp (40% of control mean); stable across extensive and intensive margins and to extensive robustness checks
- **Dynasty matters:** focusing only on parents understates intergenerational mental health persistence by 42%; social (not just genetic) transmission is implicated
- **Mental $\approx$ physical:** MH and physical health intergenerational associations are broadly similar — mental health is not uniquely persistent or uniquely genetic
- **Policy can work:** a low-touch Norwegian pilot reduced the parent-child MH association by 39%, with largest effects for youngest children
- **But equity matters:** gains accrued primarily to children of college-educated parents; the pilot may have increased inequality within the treated generation

Policy Implications and Future Directions

- **Implication 1:** Standard program evaluations that stop at the directly treated generation understate the returns to mental health investments
- **Implication 2:** Light-touch, early-childhood interventions targeting children of MH-diagnosed parents can be cost-effective — the 2010 national law scaled this approach
- **Implication 3:** Accounting for extended family links is important for understanding the *full* burden of mental illness on society
- **Open questions for future research:**
 - Causal mechanisms: exogenous shocks to parental MH (e.g., job loss, health shocks) on child outcomes
 - Do the gains from the program extend beyond mental health to labor market outcomes?
 - Optimal design of interventions that reduce average persistence *and* inequality simultaneously